FIELD GUIDE

ProvenanceOS™

A simple phone-first workflow for recording Andamooka specimens.

01	REGISTER Give the specimen an identity.
02	CAPTURE Shoot the evidence.
03	MEASURE Record dimensions where justified.
04	XRF Attach the original Vanta result.
05	COMPLETE Review. Preserve. Move on.

Mat — just shoot away.
The system will handle the rest.

What ProvenanceOS™ is doing

ProvenanceOS™ gives every physical specimen a permanent identity. Everything collected about that specimen is attached to that identity: photographs, fluorescence/afterglow evidence, measurements, XRF results and later observations.

Before you start

• Use the phone browser to open the ProvenanceOS™ field app.
• Make sure you are signed in and have internet access before starting a session.
• Have specimen bags/labels or another physical labelling method ready.
• Keep the Vanta export file intact. Do not rename, edit or resave the original export before it is imported.

Specimen classes

CODE	USE WHEN	MEANING
AHM	Andamooka Hard Matrix	Hard matrix material
ACC	Andamooka Coloured Concrete	Coloured concrete / matrix material
ACX	Andamooka Crystal Opal	Crystal opal
OTH	Unknown / Other	Use when classification is uncertain

Important: when you are unsure, use OTH. The permanent specimen ID should not be changed just because the material is later reinterpreted.

The field workflow

1. Find a specimen Take the physical stone/bag into the workflow before moving on.

2. Register it Tap + Register specimen. Choose AHM, ACC, ACX or OTH. Add location and optional notes/weight.

3. Wait for the ID ProvenanceOS™ assigns the permanent number. Do not type or guess the number.

4. Label the physical specimen Write or attach that exact ID to the stone, bag or tray.

5. Capture visible evidence Take the normal visible-light specimen photograph. Keep the original capture unchanged.

6. Capture special evidence When useful, add fluorescence and/or phosphorescence photos or video.

7. Measurement / dimensions Use the camera measurement tool when a known reference is in the same image plane.

8. Attach XRF When a Vanta result is available, attach/import the original export and match its reading to the specimen ID.

9. Finish Check the specimen record. It should show identity, evidence and available XRF/measurement data.

Keep the original export

The raw Vanta export is evidence. Do not edit the file before import.

1	EXPORT	Move the original Vanta CSV from the instrument to the device or computer used for import.
2	PRESERVE	Keep the file exactly as exported. ProvenanceOS™ stores the original file and a hash for traceability.
3	IMPORT	Use XRF Inbox / Import Vanta CSV.
4	MATCH	An exact ProvenanceOS™ specimen ID is attached to that specimen. Unmatched or conflicting rows are held for review rather than guessed.
5	REVIEW	Check the import result before assuming all readings have attached correctly.
6	INTERPRET	Values such as <LOD remain below-detection observations; they are not converted into zero.

Important: missing units are not silently assumed when the source does not provide them. Review the import warning if units are uncertain.

When something goes wrong

SITUATION	WHAT TO DO
Not sure what class a stone is	Use OTH. Make a note about the uncertainty.
Accidentally start a second record	Stop. Do not invent a number or try to merge records in the field. Keep the physical specimen identifiable and flag the duplicate for review.
A photo fails to upload	Keep the specimen ID written on the physical item and retry when connected. Do not create a new specimen just because a photo failed.
An XRF reading does not match	Do not guess. Leave it unmatched for review.
The Vanta file has <LOD / ND	Leave it as a below-detection result.
Unsure whether a value is ppm	Do not alter the value. Check the source/export and review the import warning.
Internet connection drops	Do not assume the record has saved. Keep the physical ID/label and retry when connected.

What not to do

- Do not open or edit the Master Excel workbook in the field.
- Do not create specimen IDs manually.
- Do not rename or alter original Vanta exports before import.
- Do not overwrite an earlier result because a later result looks better.
- Do not convert <LOD to zero.
- Do not guess a measurement, sample match or chemistry unit when the evidence is unclear.

03 / THE RECORD

Preserve the evidence. Flag the uncertainty.

The system is designed to keep original evidence intact and make uncertainty visible rather than silently cleaning it up.

End-of-day check

✓ Every physical specimen has a permanent ProvenanceOS™ ID.

✓ Every labelled specimen can be matched to its digital record.

✓ Important visible colour / UV / afterglow evidence has been captured.

✓ Any available XRF files are preserved and imported or clearly marked for later review.

✓ Anything uncertain has a note or review flag rather than a guess.

Scientific defensibility

• Original Vanta CSV files are stored in a private bucket and hashed with SHA-256.

• The exact parsed Vanta row is retained in analyses.raw.

• Exact specimen-ID matching is recorded as the match method with full confidence; non-exact rows are not silently attached.

• Units that were absent from the export are stored as UNVERIFIED rather than silently treated as ppm.

• <LOD, ND and related values remain below-detection flags and are never converted to zero by the application.

• Original media are immutable evidence; thumbnails and masks are derivatives.

• Derived camera measurements retain reference pixels, specimen pixels and source image ID.

• IDs are unique and concurrency-safe but intentionally not gap-free.

Vanta setup

Configure the Vanta export / Auto Export so the sample identifier is written into the export row, preferably through the Vanta Notes/Sample field. The exact specimen ID is the authoritative link between the instrument result and the physical specimen.

Treatment record

For treated matrix, record the treatment independently from the XRF measurement. An observed S result is an elemental observation; it is not a direct measurement of treatment dose. Record the treatment process and whatever inputs are actually known, then compare treated/untreated analytical observations without rewriting either record.

THE GOLDEN RULE

When in doubt: preserve the original evidence, keep the specimen identifiable, and flag the uncertainty.

We will never make the record look cleaner by guessing.

Australian Provenance Project • ProvenanceOS™ • Field Manual • Mat • v1.0